Explaining NCAA Tournament Seeding: Metrics, Decisions, and Deviations

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Abstract

NCAA tournament seeding determines bracket structure and opponent difficulty for every team in the field, yet the selection committee's decision-making process is not fully transparent. Using team-level data from Barttorvik's database spanning 2008 to 2024, we build a baseline expected-seed model using publicly available pre-tournament metrics and use the residuals from that model to investigate systematic sources of deviation in committee behavior.

We find that seeding is largely explainable from public data, with 86.3% of teams predicted within two seed lines. Wins Above Bubble (WAB) is the dominant predictor, roughly 1.4 times more influential than Barthag and 6 times more influential than efficiency metrics, suggesting the committee weights accomplished résumé over underlying team strength. We then apply this framework to three specific questions. Conference tournament performance provides no credible seed boost once regular season metrics are controlled for, across all conference tiers. Three power conferences (SEC, Big East, Big 12) are credibly over-seeded by roughly half a seed line each, while mid-majors show a weaker tendency toward under-seeding. Finally, the committee's metric weighting appears largely stable before and after the introduction of the NET ranking system in 2019, with limited evidence of any broad shift.

Overall, tournament seeding is mostly metric-driven and reproducible from public data, but a small and identifiable bias toward power conferences exists, and résumé quality matters substantially more than efficiency in how seeds get assigned.

Introduction

Every March, the NCAA tournament selection committee assigns seeds to the 68 teams in the field, determining the structure of the bracket and heavily influencing tournament outcomes. A team's seed affects matchup difficulty, advancement probability, and overall expectations, making seeding one of the most important decisions in college basketball. Despite this, the committee's decision-making process is not fully transparent, leaving teams, analysts, and fans to infer how different performance metrics are actually being weighted.

Most existing work focuses on predicting game outcomes rather than explaining how seeds are assigned. However, this misses a key step in the process: seeding determines the environment in which those games are played. This project addresses that gap by asking a central question: to what extent is NCAA tournament seeding driven by objective performance metrics, and where does the committee systematically deviate from those metrics?

This question is interesting because it sits at the intersection of data and human decision-making. If seeding is almost entirely explained by metrics, then the process is largely objective and reproducible. If not, then the differences reveal where subjective judgment, bias, or additional context plays a role. Understanding this distinction provides insight into how the committee actually evaluates teams and whether certain factors are consistently over- or under-weighted.

To answer this, we build a baseline expected-seed model using pre-tournament data and compare predicted seeds to actual outcomes. We then use these differences to analyze four key areas: (1) which metrics best explain seeding decisions, (2) whether conference tournament performance provides additional influence, (3) whether certain conferences are systematically over- or under-seeded, and (4) whether the weighting of metrics changed after the introduction of the NET ranking system in 2019.

Across all analyses, we find that NCAA tournament seeding is largely explainable using publicly available metrics, with most teams predicted within two seed lines. However, not all metrics matter equally. Resume-based measures,

particularly Wins Above Bubble (WAB), are the dominant driver of seeding decisions, while efficiency metrics provide smaller, secondary contributions. Beyond this, we find little evidence that conference tournament performance significantly impacts seeding once team quality is accounted for, and only modest, consistent conference-level biases. Finally, we find limited evidence of a major shift in committee behavior after the introduction of NET, suggesting that the overall evaluation framework has remained relatively stable over time.

Overall, this project provides a clear, data-driven framework for understanding NCAA tournament seeding and identifies where the committee's decisions align with, and diverge from, objective performance metrics.

Data

The data used in this project comes from historical NCAA men's basketball team-season records available through the cbbdata R package, which aggregates data from Barttorvik's publicly available database (https: //cbbdata.aweatherman.com/). This dataset provides detailed team performance metrics, resume statistics, and tournament outcomes across multiple seasons.

The unit of analysis is the team-season. We restrict the dataset to teams that made the NCAA tournament between2008 and 2024, excluding 2020 when the tournament was canceled. All predictor variables are measured as of Selection Sunday to ensure that the analysis only uses information that was available to the committee at the time seeding decisions were made.

The primary outcome variable is NCAA tournament seed, which takes integer values from 1 to 16. This represents the committee's final decision for each team and serves as the response variable across all models in the project.

We construct a set of predictors that capture different dimensions of team performance:

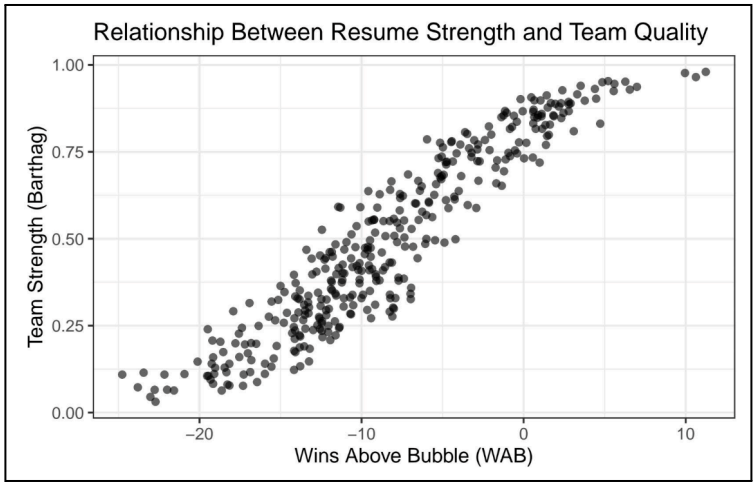
- **Efficiency-based metrics (team quality):**
  - Adjusted offensive efficiency
  - Adjusted defensive efficiency
  - Adjusted efficiency margin
  - Barthag rating
- **Resume-based metrics (team accomplishments):**
  - Wins Above Bubble (WAB)
  - Strength of schedule
  - Quadrant-based results (e.g., Quad 1 wins)
- **Contextual variables:**
  - Conference affiliation
  - Conference tournament performance (e.g., wins or advancement)
  - Era indicator for pre-2019 vs. post-2019 (NET introduction)

These variables allow us to separate different explanations for seeding decisions, including overall team quality, résumé strength, and contextual factors that may influence committee behavior.

Several preprocessing steps are applied before analysis. We restrict the dataset to tournament teams, standardize team and conference naming across seasons, and remove observations with missing or inconsistent values. Continuous predictors are scaled to have mean zero and standard deviation one for comparability across models. Additional derived variables are created to support specific sub-analyses, including conference tournament performance indicators and a NET-era indicator.

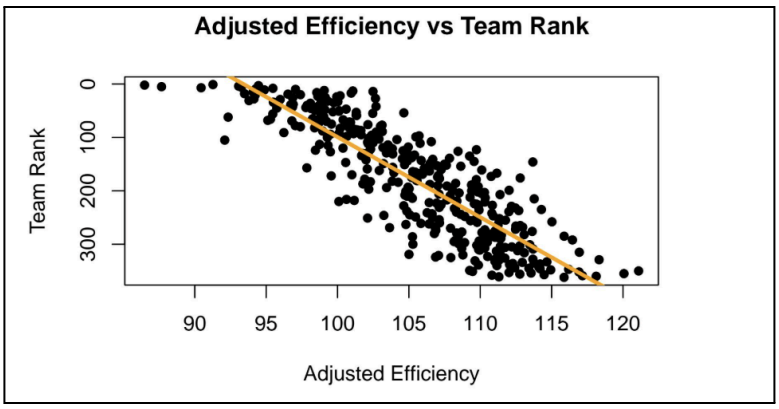
Exploratory Data Analysis

We begin by exploring how key metrics relate to tournament seeding and how these relationships vary across contexts. The goal of this analysis is to validate that the selected predictors capture meaningful differences between teams and to identify patterns that motivate the modeling approach used throughout the project.



EDA 1: *Team Strength vs. WAB*

This plot examines the relationship between Wins Above Bubble (WAB), a resume-based metric, and Barthag, which measures overall team quality. As expected, there is a strong positive relationship. Teams with higher WAB tend to also have higher Barthag values, since better teams generally win more games. However, the relationship is not perfectly linear, and there is noticeable spread, especially in the middle range. This shows that resume-based metrics and efficiency-based metrics are related but capture different aspects of team performance. This distinction is important for our project, because it suggests that the selection committee may be weighing "what a team accomplished" differently from "how strong a team looks on paper." This motivates our modeling approach, where we explicitly separate resume and efficiency metrics to evaluate their relative importance in determining NCAA tournament seeding.



EDA 2: *Team rank vs Adjusted Defensive Efficiency*

The scatterplot above shows the relationship between adjusted defensive efficiency (points allowed per 100 possessions) and overall team ranking, with the y-axis inverted so that stronger teams appear toward the top. Teams with better defensive efficiency tend to be ranked higher, but the substantial scatter around the fitted line shows that defense alone does not determine team strength, indicating that other variables may also play important roles. This motivates our use of multi-variable models in the following sections.

Emilia Brusseau: Baseline Expected-Seed Model

Methods

The goal of this analysis is to figure out which publicly available team metrics best explain how the NCAA selection committee assigns tournament seeds. If seeding is mostly driven by objective performance metrics, then a model built on those metrics should be able to predict seeds pretty accurately. When the model is off, those differences highlight where the committee is relying on something beyond the numbers and those residuals are what the rest of our project focuses on.

The outcome variable is NCAA tournament seed (1 through 16), which I treat as a 16-category unordered variable and model using multinomial logistic regression with nnet::multinom(). I'm not using an ordinal model because that would assume the gaps between seeds are evenly spaced, which doesn't really make sense here. The difference between a 1 and 2 seed is not the same as between an 8 and 9 seed in terms of matchup difficulty or tournament path. The multinomial model avoids that by estimating separate log-odds for each seed relative to a reference category (seed 1), without forcing equal spacing. The model assumes these log-odds are a linear function of the predictors and that each team-season is independent. I standardize all predictors to have mean 0 and standard deviation 1 so the coefficients are directly comparable across variables.

To break down what actually drives seeding, I compare three nested models:

- **M1 (Efficiency only):** adjusted offensive efficiency, adjusted defensive efficiency, and Barthag (a power rating that reflects expected win probability against an average Division I team). I leave out adjusted efficiency margin since it's basically a combination of offense and defense and would cause collinearity issues.
- **M2 (Resume only):** Wins Above Bubble (WAB), which captures how a team performed relative to a bubble-level team given the same schedule, and non-conference elite strength of schedule.
- **M3 (Combined):** all five predictors together.

I fit all models using a stratified 80/20 train/test split (stratified by seed so every seed appears in both sets). I evaluate performance on the holdout set using AIC (on the training data), mean absolute error in seed units, exact seed accuracy, within-2-seeds accuracy, and multinomial log-loss.

To quantify uncertainty, I use a nonparametric bootstrap with 200 resamples of the training data. This gives 95% confidence intervals for each predictor's average absolute coefficient, which I use to compare their overall importance in the model.

Results

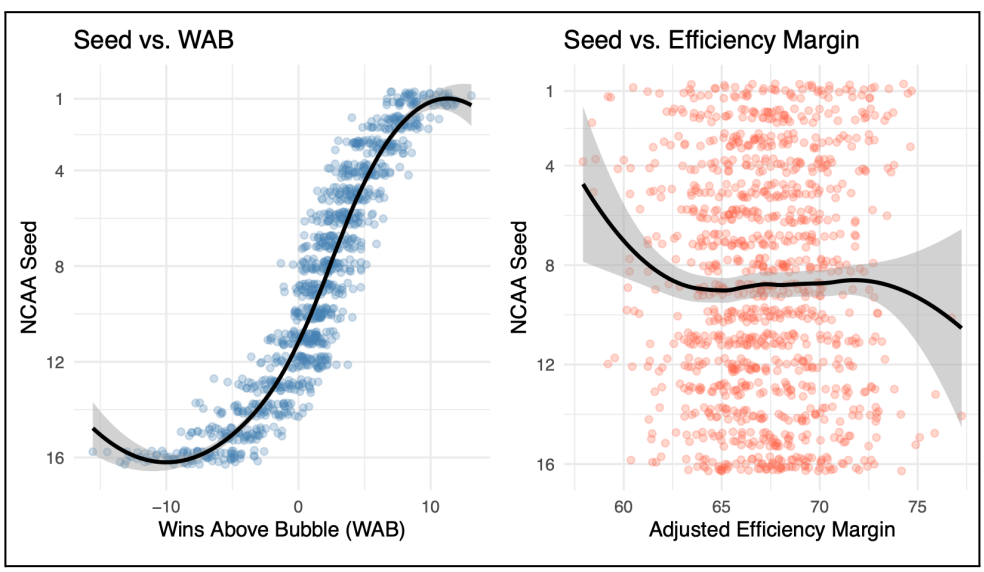


Figure 1: *WAB (left) shows a tighter, more linear relationship with seed than adjusted efficiency margin (right), suggesting the committee weights resume-based metrics heavily in seeding decisions. Each point is one tournament team-season, 2008-2024.*

Model	AIC	Mean Abs Error	Exact Accuracy	Within-2 Acc	Log-Loss
M1: Efficiency Only	3165.8	1.527	0.278	0.785	1.850
M2: Resume Only	2863.3	1.195	0.298	0.844	1.646
M3: Combined	**2695.7**	**1.137**	**0.332**	**0.863**	**1.608**

Table 1: *Holdout set performance across three models. M3 outperforms both single-category models on every metric, showing that efficiency and resume metrics each carry independent signals for seeding. M2 outperforms M1 by a larger margin than M3 outperforms M2, showing that resume metrics are the dominant driver.*

The combined model (M3) correctly predicts the exact seed for 33.2% of teams in the holdout set and is within two seeds for 86.3% of teams, with a mean absolute error of 1.14 seed lines. This outperforms both M1 (efficiency only) and M2 (resume only), which suggests that both types of metrics matter and are capturing different pieces of the selection process. That said, the improvement from M1 to M2 is much larger than from M2 to M3 (AIC drops by 302 vs. 167), which points to resume-based metrics as the main thing the committee is using. Efficiency metrics still add value, but more secondary rather than the primary driver.

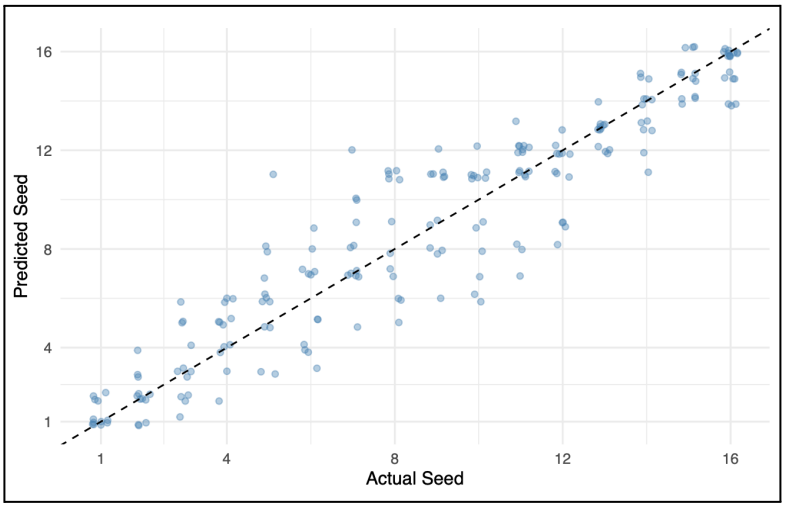


Figure 2: *Predicted versus actual seed for M3 on the holdout set. The dashed line represents a perfect prediction. The model tracks the committee well across the full range of seeds, with most errors confined to adjacent seed lines. The largest deviations occur in the middle of the seed distribution (seeds 7-11), where the committee has the most discretion.*

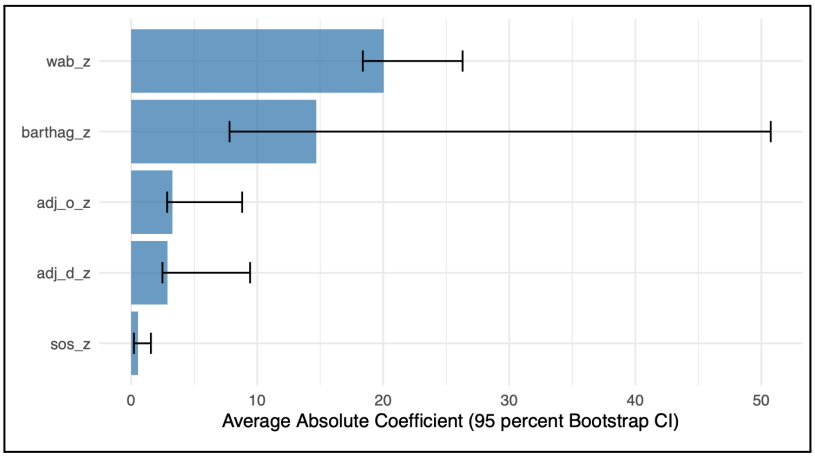


Figure 3: *Average absolute log-odds coefficient by predictor in M3, with 95 percent bootstrap confidence intervals (B = 200). WAB dominates by a wide margin, with an average coefficient around 1.4 times larger than Barthag and 6 times larger than the efficiency components. The non-overlapping intervals for WAB and Barthag versus adj_o, adj_d, and so prove that this ordering is stable across resamples.*

WAB is clearly the most influential predictor in the model, with an average absolute coefficient of 20.0. That's noticeably larger than Barthag (14.7), and way above the efficiency metrics (3.3 for adjusted offense, 2.9 for adjusted defense) and strength of schedule (0.5). The bootstrap confidence intervals back this up.

WAB's interval (18.4, 26.3) doesn't overlap with the efficiency metrics at all, and even though Barthag's interval (7.8, 50.8) is wide, it still sits well above the efficiency variables.

That wide interval for Barthag is actually expected because it reflects some instability in how the model splits credit between Barthag and WAB across different resamples, since they're pretty correlated. But even with that, both consistently matter much more than the pure efficiency metrics.

Overall, this points to a pretty clear takeaway: the committee seems to care more about what a team actually did against its schedule (WAB) than how "good" it looks in efficiency terms. Strength of schedule doesn't add much once WAB is in the model, which makes sense since WAB already builds that in.

The combined model explains about 86.3% of teams within two seed lines, leaving roughly 14% of cases where the committee deviates from what the metrics would predict. Those gaps in which those teams are seeded noticeably higher or lower than expected are what the rest of our project focuses on, including things like conference tournament effects, conference-level bias, and changes after the introduction of NET rankings in 2019.

Sam Sclaver: Conference Tournament Seed Boost

Methods

The goal of this analysis is to determine whether conference tournament performance provides a systematic seed boost beyond what a team's regular season metrics already predict, and whether this effect varies by conference tier. The outcome variable is the seed residual from the baseline multinomial model (actual seed minus predicted seed), where a positive residual indicates the committee seeded the team worse than predicted and a negative residual indicates the committee was more generous than expected.

The key predictor is the number of conference tournament wins a team accumulated, derived from game-level data in cbbdata by filtering to conference tournament games and counting wins per team per season. All baseline model predictors are measured as of Selection Sunday, meaning they already reflect any metric changes from conference tournament performance. This means the seed residual isolates whether the committee rewards conference tournament wins beyond what the updated metrics already capture.

To examine whether the per-win effect differs across conference strength, I classified conferences into three tiers based on historical at-large bid rates from 2008 to 2024. At-large bids are awarded to teams that did not win their conference tournament but are deemed worthy by the committee, so a conference's at-large rate reflects how consistently its teams are viewed as tournament-caliber beyond just the automatic bid winner. Power conferences (ACC, Big Ten, Big 12, SEC, Big East, Pac-10/12) have at-large rates above 75%, mid-major conferences (A-10, Mountain West, AAC, WCC) range from 48 to 68%, and low-major conferences have at-large rates below 20%. The Ivy League is classified as low-major despite a 47% at-large rate because this reflects only 15 total tournament appearances and 7 at-large bids across 17 seasons, both well below any conference classified as mid-major.

Before fitting a formal model, I conducted an initial analysis using the baseline expected seed model to visually assess the raw relationship between conference tournament performance and seed residuals. I calculated the mean seed residual for each exact number of conference tournament wins, grouping the teams by their respective conference tier. To quantify uncertainty in these initial estimates, I computed 95% confidence intervals around each group mean using the standard error of the mean under a normal approximation. This initial step provides a baseline view of the data structure before introducing hierarchical shrinkage.

The primary model is a Bayesian hierarchical model fit using :

$$seed_residual_i = \beta_0 + \beta_1 \cdot conf_t_wins_i + u_{0,j} + u_{1,j} \cdot conf_t_wins_i + \epsilon_i$$

$$u_{0,j}, u_{1,j} \sim MVN(0, \Sigma), \quad \epsilon_i \sim Normal\left(0, \sigma^2\right)$$

where j indexes conference tier and $u_{0,j}$ and $u_{1,j}$ are tier-level random intercepts and slopes with partial pooling. Priors are weakly informative: $Normal(0, 3)$ on fixed effects and $Exponential(1)$ on standard deviation parameters. The model was fit with 4 chains and 4000 iterations per chain. Partial pooling is appropriate because the per-win effect likely varies across tiers but each tier has limited data, and independent estimation would produce noisy estimates. As a comparison, I fit a simple linear regression without tier structure and a frequentist interaction model with conference tournament wins interacted with tier. Model fit is validated through posterior predictive checks.

Results

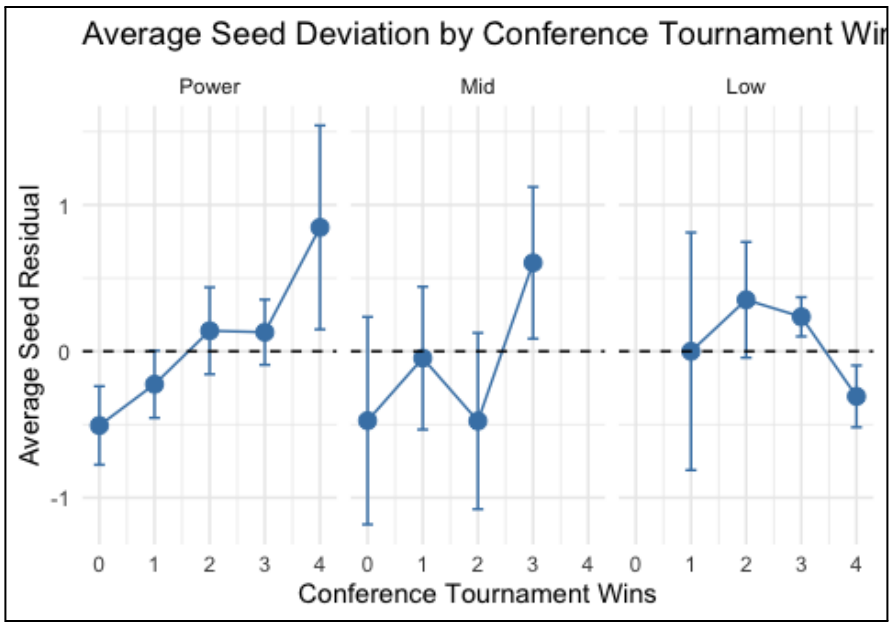


Figure 4: *Average seed residual from baseline expected seed model by conference tournament wins, faceted by conference tier. A positive residual indicates the committee seeded the team worse than the baseline model predicted. Error bars represent 95 percent confidence intervals. Groups with fewer than 10 observations are excluded.*

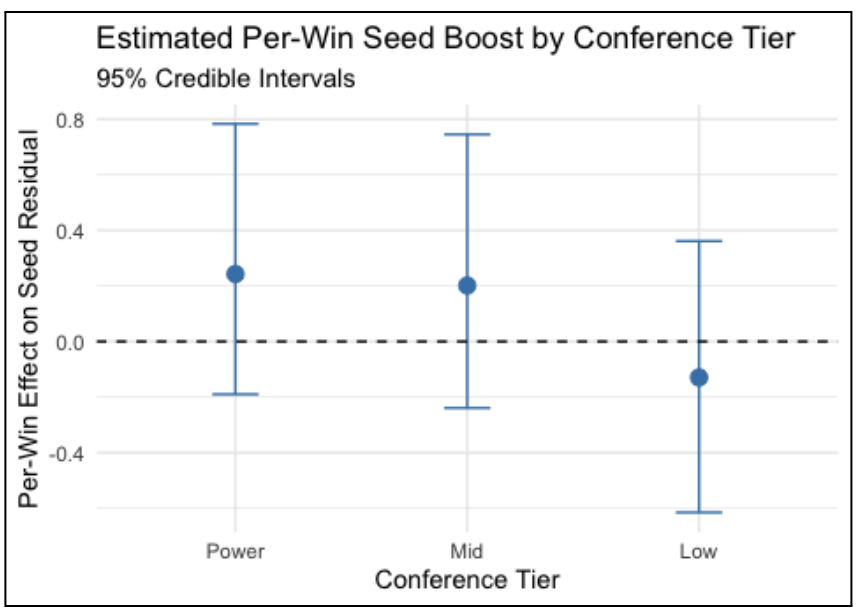


Figure 5: *Estimated per-win seed boost from the Bayesian hierarchical model by conference tier with 95 percent credible intervals. All three tiers have intervals crossing zero, indicating no credible per-win effect in any tier.*

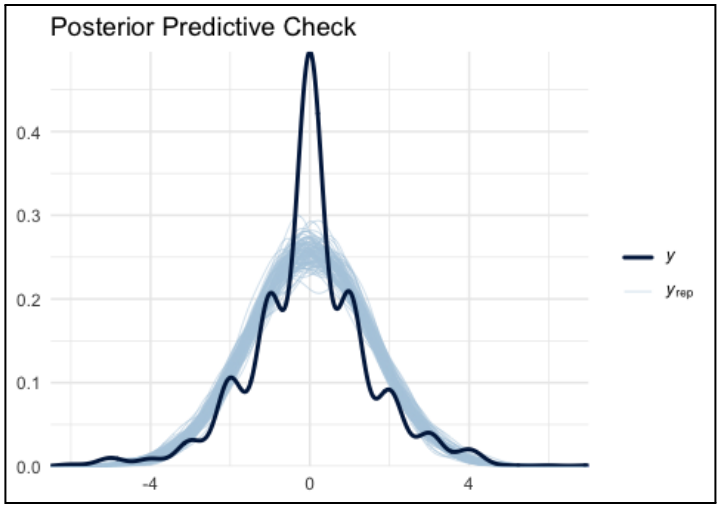


Figure 6: *Posterior predictive check comparing observed seed residuals (dark line) to 100 simulated datasets from the posterior (light lines). The model captures the overall spread but does not replicate the discrete integer structure of the observed residuals.*

An initial analysis using the baseline expected seed model reveals no clear pattern suggesting conference tournament wins improve seeding (Figure 4). In power conferences, teams with zero wins have the most negative residuals, with residuals increasing as wins accumulate. Mid-major conferences show a similar pattern with wider uncertainty, while low-major conferences trend slightly negative as wins increase.

The Bayesian hierarchical model estimates an overall per-win effect of 0.092 seed lines (95% credible interval: [-0.429, 0.526]), with the credible interval comfortably containing zero. Tier-specific slopes are 0.243 for power, 0.201 for mid-major, and -0.129 for low-major (Figure 5), with all three credible intervals crossing zero.

The divergence between low-major and the other two tiers is not surprising when considering how teams from these conferences reach the tournament. Power and some mid-major conference teams frequently earn at-large bids based on their regular season resume alone, meaning conference tournament performance is additional information on top of an already established profile. Low-major teams, by contrast, pretty much exclusively reach the tournament by winning their conference tournament, so their conference tournament wins are not new information to the committee but rather the prerequisite for being in the field at all. These are teams that don't have the resume to get in the tournament without an auto-bid (winning their conference tournament), which explains why the low-major per-win slope trends are slightly negative rather than positive.

To validate, I compared against two frequentist models. The simple linear regression estimates a per-win effect of 0.192 ($p < 0.001$), but this is misleading because it pools all tiers. The interaction model separates tiers, with slopes of 0.257 for power, 0.23 for mid-major, and -0.229 for low-major, with the low-major interaction significant at $p < 0.001$. The Bayesian model produces similar tier-specific slopes but shrinks all estimates toward zero through partial pooling. The frequentist interaction model explains only 4% of variance in seed residuals, confirming that conference tournament performance accounts for a very small portion of seeding deviations.

Posterior predictive checks (Figure 6) confirm the model captures the overall spread of residuals, though the Gaussian likelihood produces smooth simulated distributions that do not replicate the discrete integer structure of the observed data. The model was fit with adapt_delta = 0.99, with 22 divergent transitions remaining out of 8000 draws, attributable to estimating hierarchical variance components with only three grouping levels.

The central finding is that the NCAA selection committee does not appear to systematically reward or penalize conference tournament performance beyond what regular season metrics already capture. While conference tournament results are known to be one of many factors the committee considers, this analysis suggests their influence on seeding is minimal once team quality is controlled for. This holds across all three conference tiers and is robust to model specification, with both the Bayesian hierarchical model and the frequentist interaction model pointing to the same conclusion. The committee appears to evaluate teams primarily on their full body of work as captured by metrics like WAB and adjusted efficiency, rather than giving additional weight to late-season conference tournament runs.

Shiv Singhal: Conference Bias

Methods

This analysis aims to figure out whether the selection committee systematically over-seeds or under-seeds certain conferences (when we control for team quality). The baseline model already explains ~86% of teams within two seed lines using only Barttorvik resume and efficiency metrics, so anything that conference affiliation adds would be coming from how the committee treats the conference itself, not how good the teams in it actually are.

Our response variable is the seed residual, which is actual seed minus expected seed from the combined multinomial model (M3), where the expected seed is computed as $\sum_{k=1}^{16} k \cdot Pr(seed_i = k \mid x_i)$. A negative residual means the team was seeded better than its metrics warranted (over-seeded), and a positive residual means the team was seeded worse than its metrics warranted (under-seeded). Focusing on residuals lets us separate committee behavior from team performance. If a certain conference's teams happen to have great metrics, the baseline captures that rather than the residual.

The main complication is sample size. Across 2008-2024 (excluding 2020), the major conferences (ACC, Big Ten, Big 12, SEC, Big East, Pac 12) contribute on the order of 90-120 tournament team-seasons each, while small one-bid conferences contribute fewer than 20. A simple per-conference mean residual would be very noisy for the small leagues and could easily produce a "biggest bias" estimate.

To account for the sample size issue, we fit a **Bayesian hierarchical model** with conference-level random intercepts:

$$r_i = \beta_0 + \alpha_{c[i]} + \varepsilon_i, \qquad \alpha_c \sim \mathcal{N}(0, \tau^2), \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

Here, r_i is the seed residual for team-season i, $c[i]$ is its conference, and α_c is the conference effect estimated with partial pooling. The priors are $\beta_0 \sim N(0, 1)$, $\sigma \sim$ Half-$N(0, 2)$, and $\tau \sim$ Half-$N(0, 1)$. We chose Half-Normal distributions to ensure that scale parameters are positive. As stated above, $\alpha_c > 0$ means conference c is on average **under-seeded** and $\alpha_c < 0$ means that conference c is **over-seeded**. The full posterior of α_c is what we will use for point estimates and probabilities.

Partial pooling is an effective method choice because conferences with only a handful of tournament teams have huge sampling variance in their raw mean residuals, and partial pooling shrinks those estimates toward the overall mean by an amount that depends on how much information the conference actually provides. That directly controls the false-positive rate as smaller conferences can't claim a large "bias" effect unless the data are very consistent.

The model assumes residuals are approximately normal given the conference effect, residuals from different team-seasons within a conference are conditionally independent. The normality assumption is reasonable because the expected seed is computed as a weighted average over 16 categories, which smooths out the discreteness. We will confirm this assumption with posterior predictive checks on the residual distribution.
We compare the hierarchical model against three simpler alternatives that all target individual conference mean residual, but handle pooling differently:

1. **Complete Pooling**: a single intercept, $r_i = \beta_0 + \varepsilon_i$. This forces every conference effect to be zero and serves as a null benchmark.
2. **No Pooling (Fixed Effects)**: $r_i = \beta_{c[i]} + \varepsilon_i$. This has a separate intercept for each conference and no shrinkage.
3. **Bootstrap of Individual Conference Mean Residuals**. For each conference, we resample its team-seasons with replacement $B = 2000$ times and compute a 95% percentile interval. This is a nonparametric residual-summary approach and confirms the parametric assumptions in the hierarchical model.

We evaluate the hierarchical model against the no-pooling and complete-pooling versions using leave-one-out cross-validation (LOOCV) on the Bayesian fits, and against the bootstrap by visually overlaying intervals in the conference forest plot. The expectation is that LOOCV will prefer the hierarchical model if there is real cross-conference variation but not to the extent that no-pooling is justified, and that the hierarchical intervals will be tighter than bootstrap intervals for small conferences (because of shrinkage) but very similar for large ones.
For the hierarchical model, uncertainty is captured by the posterior. For each conference, we report

the posterior median and 95% credible interval for α_c, and also the posterior probability that $\alpha_c > 0$ (under-seeded) and $\alpha_c < 0$ (over-seeded). We treat Pr(*under-seeded*) > 0.95 or Pr(over-seeded) < 0.95 as "credible evidence of bias" for that conference. We run 4 chains, 2,000 iterations each (1,000 as warmup), and check convergence with $\widehat{R}$ and bulk/tail effective sample sizes above 400.

For the no-pooling and complete-pooling Bayesian models we report the same posterior summaries. For the bootstrap approach we report the percentile interval and the bootstrap-based proportion of resamples with mean residual greater than zero. We also run posterior predictive checks by simulating new residual datasets from the fitted hierarchical model and comparing the simulated distribution of residuals (overall and by conference) to the observed one.

Results

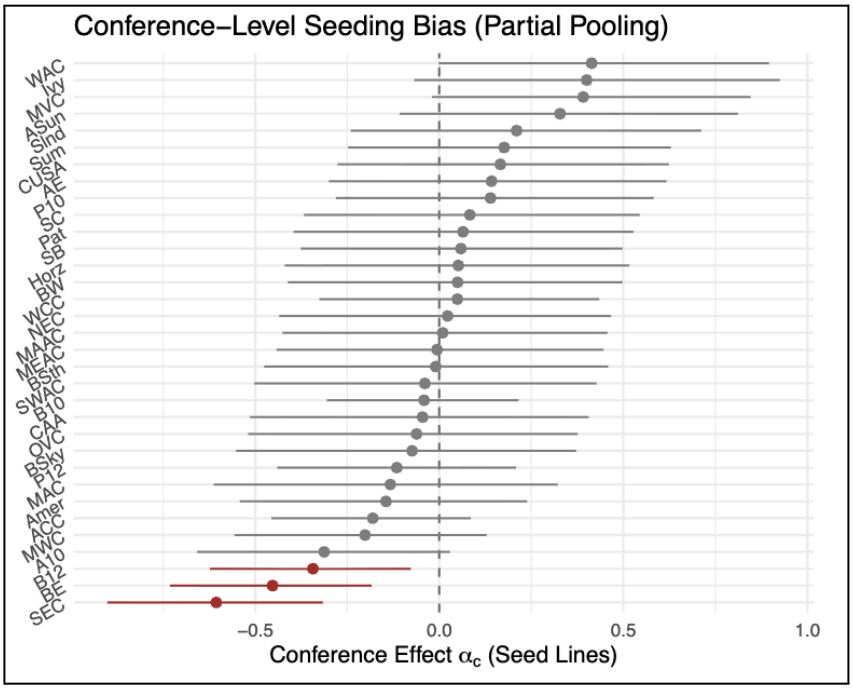


Figure 7: *Posterior Conference Effects from the Bayesian Hierarchical Model. Conferences Ordered by Posterior Median. Sample Size n is Number of Tournament Team-Seasons (2008–2024)*

Conference	n	Median (Seeds)	95% CrI	P(over)	P(under)
SEC	84	-0.61	[-0.90, -0.32]	1.00	0.00
BE	101	-0.45	[-0.73, -0.18]	1.00	0.00
WAC	17	+0.41	[-0.00, +0.90]	0.03	0.97
Ivy	15	+0.40	[-0.07, +0.92]	0.05	0.95
MVC	21	+0.39	[-0.02, +0.85]	0.03	0.97
B12	103	-0.34	[-0.62, -0.08]	0.99	0.01
ASun	16	+0.33	[-0.11, +0.81]	0.07	0.93
A10	47	-0.31	[-0.66, +0.03]	0.96	0.04

Table 2: *Top 8 Conferences by Absolute Posterior Median Effect. P(over) and P(under) are the Probabilities of Over-Seeding and Under-Seeding, Respectively.*

Three conferences show credible bias. The SEC, Big East, and Big 12 are over-seeded by roughly half a seed line each, with posterior probabilities of over-seeding at 0.99 or above. A handful of smaller conferences (WAC, Ivy, MVC, ASun) lean toward being under-seeded by a similar magnitude, but their 95% credible intervals all include zero. The pattern hints that mid-majors get slightly less benefit of the doubt than their metrics would predict, but from this information we can't call any single conference credibly biased. Across the board, no effect exceeds 0.61 seed lines. The committee shows real bias, but at a magnitude unlikely to change a team's bracket region in any given year.

Conference	n	Hierarchical Median	Hierarchical 95% CrI	Bootstrap Mean	Bootstrap 95% CI
SEC	84	-0.61	[-0.90, -0.32]	-0.61	[-0.90, -0.32]
BE	101	-0.45	[-0.73, -0.18]	-0.40	[-0.65, -0.17]
WAC	17	+0.41	[-0.00, +0.90]	+0.99	[+0.31, +1.90]
Ivy	15	+0.40	[-0.07, +0.92]	+1.03	[+0.36, +1.81]
MVC	21	+0.39	[-0.02, +0.85]	+0.87	[+0.26, +1.59]
B12	103	-0.34	[-0.62, -0.08]	-0.27	[-0.54, -0.01]

Table 3: *Hierarchical Posterior Intervals vs. Bootstrap Percentile Intervelas for the 6 Conferences with the Largest Absolute Hierarchical Effects.*

The bootstrap comparison makes the value of partial pooling concrete. The SEC (n = 84) shows identical hierarchical and bootstrap intervals, meaning that at that sample size pooling does nothing. WAC and Ivy (n = 17 and n = 15) tell the opposite story as bootstrapping puts both around +1.0 seed lines, while the hierarchical model shrinks them to about +0.4 with intervals that just barely touch zero. This tells us that a 17-team conference can't justify a one-seed-line effect. LOOCV favors the hierarchical fit over both alternatives on out-of-sample predictive accuracy. Complete pooling misses real variation and no pooling overfits. The hierarchical model captures the real cross-conference signal without inflating it past what 15-20 team-seasons can support.

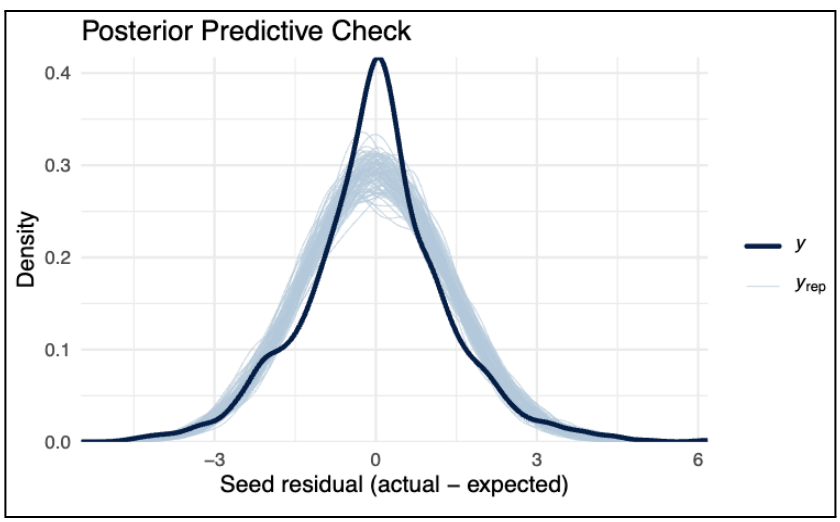


Figure 8: *Posterior Predictive Check for the Hierarchical Model.*

The posterior predictive check confirms the model reproduces the residual distribution well. The replicated densities strongly reflect the observed one across the full range, with no systematic tail mismatch, meaning the normality assumption is appropriate for conference-level inference.

To conclude this analysis on conference bias, after controlling for team quality, only three conferences (the SEC, Big East, and Big 12) are credibly over-seeded, by 0.34 to 0.61 seed lines. A handful of mid-majors lean toward being under-seeded but stop just short of credibility. The effects are small, stable across modeling choices, and survive partial-pooling shrinkage, which means they are real but limited.

Danny Weng: NET Era Shift & NCAA Tournament Seeding

Data & Sample

I pulled the Torvik team-season data through cbbdata once, saved it as a local CSV, and then used that file for the rest of this analysis. That way the report does not depend on live API calls every time it gets knitted.

The dataset includes team quality metrics along with NCAA tournament seeds by season. For my section, I keep seeded teams only, drop 2020 because there was no NCAA tournament, and create a post_net indicator for 2019 and later so I can compare the pre-NET and post-NET eras directly.

Sample used for my NET-era seeding analysis.

total_team_seasons	pre_net_team_seasons	post_net_team_seasons	seasons_used
1078	738	340	2008-2024, excluding 2020

Main variables used in my model and what they mean.

raw_name	label	meaning
barthag	Barthag	Torvik's overall team strength rating, based on expected win probability against an average Division I team on a neutral floor.
adj_o	Adjusted offensive efficiency	Points scored per 100 possessions after adjusting for opponent strength.
adj_d	Adjusted defensive efficiency	Points allowed per 100 possessions after adjusting for opponent strength.
adj_t	Adjusted tempo	Adjusted possessions per game.
wab	Wins Above Bubble	How much better or worse a team performed than a bubble-level team would be expected to perform against the same schedule.
nc_cur_sos	Current non-conference strength of schedule	How difficult the team's non-conference schedule was up to Selection Sunday.
ov_cur_sos	Current overall strength of schedule	How difficult the team's full schedule was up to Selection Sunday.

Table 4: *Dataset*

Methods

My part of the project asks a pretty specific question: did the selection committee seem to weigh the usual team-quality and resume metrics differently once the NET system showed up in 2019?

This setup is also the one that best matches the project context file and the rubric. The context markdown says my section should use a course-supported method, compare models, and quantify uncertainty for the specific NET-era question. It also recommends multinomial logistic regression as the safest choice because that method was explicitly covered in class, while ordinal regression was not clearly documented in the course materials.

Model

To look at that, I treat each seeded team-season as one observation and model final NCAA seed as a categorical outcome. I use multinomial logistic regression because seed is a 16-category response and that method lines up with the course material on multinomial models, held-out evaluation, and uncertainty-aware interpretation. The predictors I keep in the model are Barthag, adjusted offensive efficiency, adjusted defensive efficiency, adjusted tempo, Wins Above Bubble, current non-conference strength of schedule, and current overall strength of schedule. Table 2 gives a plain-English definition for each one. I standardize those variables first so the coefficient sizes are easier to compare on the same scale.

I fit four versions of the model. The first is a pooled model with the main predictors and a post_net indicator. The second is the full interaction model, which adds a post-2019 interaction for every predictor. The third is a simpler targeted interaction model that only lets the three most plausible metrics move across eras: Barthag, Wins Above Bubble, and overall strength of schedule. The last step is a pair of separate era-specific models, one fit on the pre-2019 seasons and one fit on the post-2019 seasons, just to check whether the interaction results line up with the side-by-side coefficient patterns.

Assumptions

This is still an observational model, so I am not treating any of these effects as causal. I am just asking whether the committee's seeding behavior appears to line up differently with public metrics before and after the NET era. I also assume the 2019 split is a reasonable way to mark the start of that change, even though the real transition in committee behavior could have been less clean than a single cutoff. Those assumptions are consistent with the project context file, which frames this section as a question about changes in committee weighting rather than a causal claim about why those changes happened.

Evaluation and Uncertainty

For evaluation, I use a held-out test split and compare models with multiclass log-loss first, then exact seed accuracy and within-one-seed accuracy as secondary summaries. I also repeated the train/test split several times as a stability check, since one split can be noisy with only a few post-NET seasons. That follows the course emphasis on out-of-sample model comparison and also matches the rubric guidance in the context file, which says not to rely on raw accuracy alone. For uncertainty, I report approximate 95% intervals for the interaction terms. That is important because the rubric explicitly requires uncertainty quantification, and the real question here is not whether one model can classify seeds a little better, but whether there is enough evidence to say some metrics became more or less tied to seeding after NET.

Results

The goal of this section is to test one narrow question: whether the relationship between standard team metrics and NCAA seed looks different after the committee started using NET in 2019.The analysis sample includes 1078 seeded team-seasons from 2008 through 2024, with 2020 removed because there was no tournament. That leaves 738 pre-NET observations and 340 post-NET observations.

Model Comparison

Held-out model comparison for Danny's NET-era analysis.

model	log_loss	accuracy	within_one_seed
Pooled no-interaction	1.5566	35.5%	73.2%
Pooled full interaction	1.6810	32.3%	71.4%
Pooled targeted interaction	1.5878	33.6%	72.7%

Table 5: *Held-Out Model Comparison*

On the held-out test set, the pooled no-interaction model had a log-loss of 1.557. The full interaction model was worse at 1.681, while the simpler targeted interaction model came in at 1.588. Exact seed accuracy was 35.5%, 32.3%, and 33.6%, respectively. Within-one-seed accuracy was 73.2%, 71.4%, and 72.7%. That comparison is the main test of the NET-era question. If allowing post-2019 effects really helps, one of the interaction models should outperform the stable baseline. Here, the full interaction model does not. It is worse on log-loss and slightly worse on both accuracy summaries, which suggests that a broad across-the-board shift in committee weighting is not strongly supported by out-of-sample performance. The targeted interaction model is the more realistic stress test, since it only gives extra flexibility to the metrics that look most plausible. In this sample, though, it still does not beat the no-interaction baseline.

Train-versus-test comparison for the two pooled models.

split	model	log_loss	accuracy	within_one_seed
Train	Pooled no-interaction	1.3563	45.8%	77.2%
Test	Pooled no-interaction	1.5566	35.5%	73.2%
Train	Pooled full interaction	1.2977	48.5%	78.3%
Test	Pooled full interaction	1.6810	32.3%	71.4%
Train	Pooled targeted interaction	1.3321	47.7%	77.7%
Test	Pooled targeted interaction	1.5878	33.6%	72.7%

Table 6: *Train-vs-Test Comparison*

The train-versus-test table helps explain why. On the training data, the larger interaction models do a little better than the no-interaction model, but that advantage fades or reverses on the test set. That is the pattern I would expect from mild overfitting: the larger model is flexible enough to fit extra noise in the training sample, especially with only a few post-2019 seasons, but that extra flexibility does not carry over to new observations.

Average holdout performance across 10 stratified train/test splits.

model	mean_log_loss	sd_log_loss	mean_accuracy	mean_within_one_seed
Pooled no-interaction	1.5199	0.0299	37.7%	74.6%
Pooled targeted interaction	1.5602	0.0334	37.7%	74.2%
Pooled full interaction	1.6362	0.0525	36.8%	74.0%

Table 7: *Average Holdout Performance*

The repeated-split check points in the same direction. Across 10 different stratified train/test splits, the no-interaction model has the best average log-loss at 1.52, compared with 1.636 for the full interaction model and 1.56 for the targeted interaction model. That makes me more confident that the weak NET-shift result is not just coming from one unlucky split.

Interaction Estimates and Uncertainty

Summary of which metrics shift the most in the interaction model.

metric	mean_abs_interaction	share_intervals_excluding_zero
barthag	4.248	0.0%
adj_o	4.076	0.0%
wab	2.936	0.0%
ov_cur_sos	1.902	0.0%
adj_d	1.897	0.0%
adj_t	1.388	13.3%
nc_cur_sos	0.345	0.0%

Table 8: *Metric Shift Summary*

Table 8 still matters, because even if the full interaction model does not improve overall prediction, the interaction terms can show whether a smaller number of metrics moved more than the others. Looking across seed categories, the largest average interaction shifts come from Barthag, Adjusted offensive efficiency, Wins Above Bubble. In plain terms, those are the variables whose relationship with seed changes the most between the pre-NET and post-NET eras.

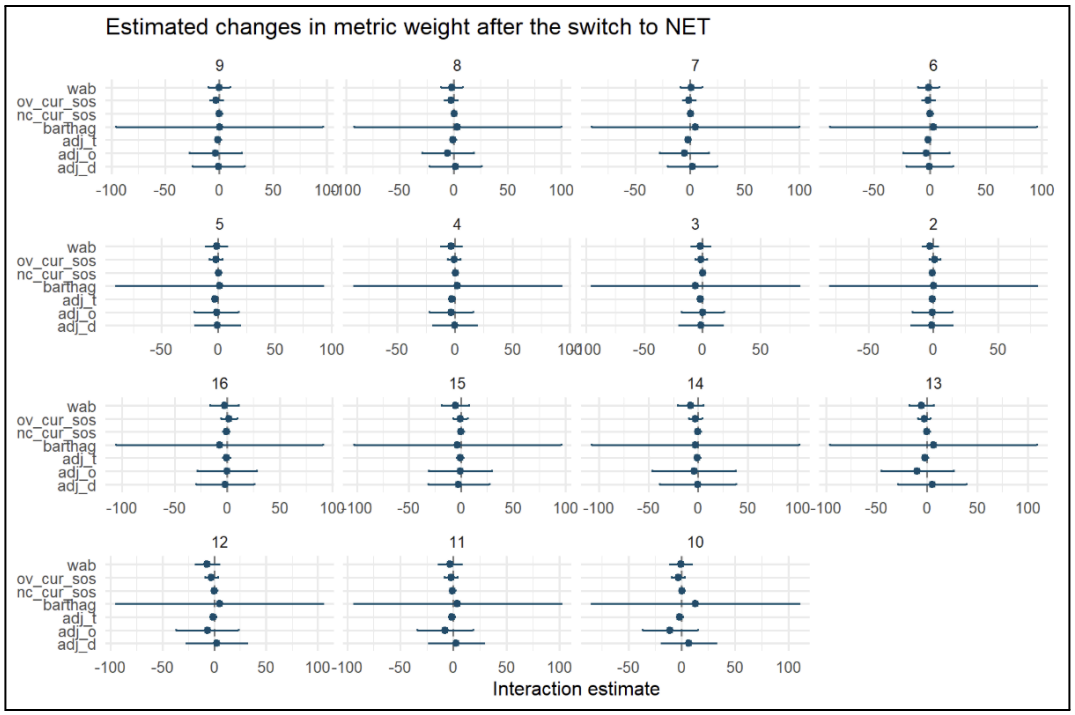


Figure 9: *Estimated NET-era interaction terms from the pooled multinomial model.*

At the same time, the interval estimates make it clear that not every apparent shift is precise. Many of the interaction bars still overlap zero, so I should be careful not to oversell the story. The safest conclusion is that the committee's weighting does not look completely identical across eras, but the evidence for a major system-wide change after NET is weak in this model. If there was a shift, it seems more likely to have been concentrated in a small subset of metrics rather than showing up clearly across the whole profile.

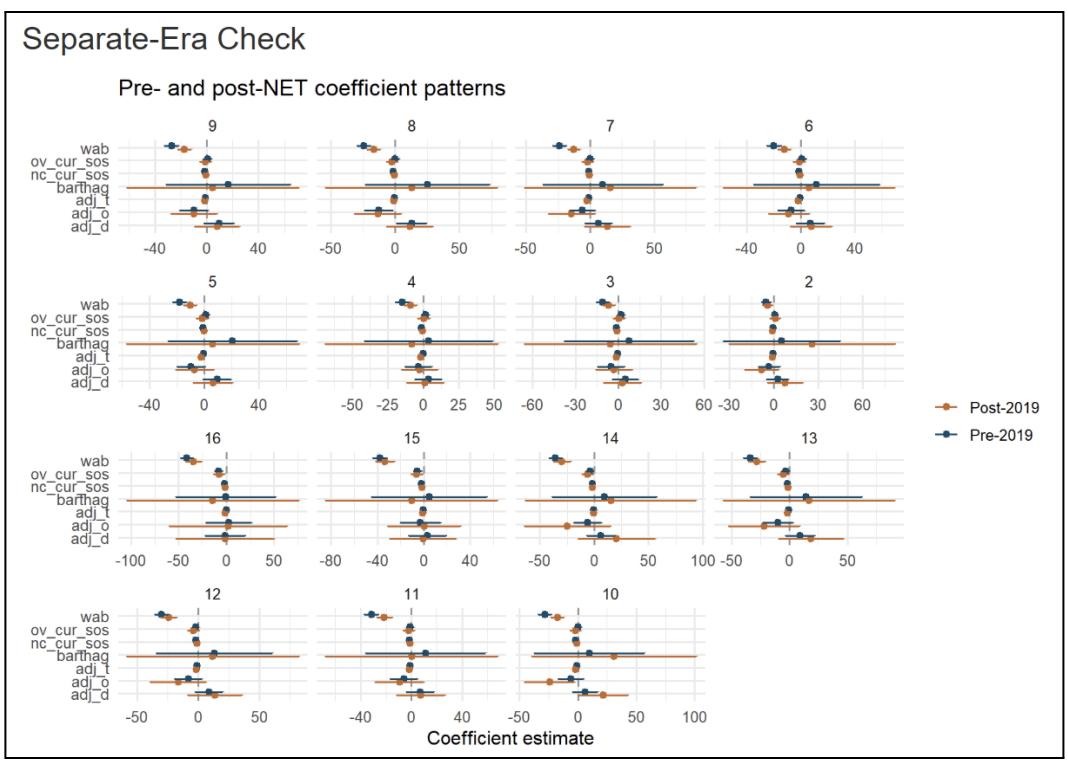


Figure 10: *Side-by-side coefficient comparison from separate pre-2019 and post-2019 multinomial models.*

Figure 10 is the main robustness check. When the separate pre-2019 and post-2019 models show differences in the same places as the interaction model, that makes those differences more believable. Overall, though, my read of this section is still pretty cautious: the data do not support a strong claim that the committee broadly changed how it seeds teams after the introduction of NET. The more defensible takeaway is that the overall seeding pattern stayed fairly similar, with only limited evidence that a few specific metrics may have shifted in importance.

The exact-seed accuracy should also be interpreted in context. This is a 16-category prediction problem, and the committee has the most discretion in the middle seed lines, so exact classification is a pretty strict standard. The no-interaction model still lands within one seed line 73.2% of the time, which is a more realistic way to judge whether the model is broadly tracking the committee.

To conclude, I do not think the right takeaway is that NET caused a big reset in how teams get seeded. Based on the held-out comparison, the repeated-split check, and the uncertainty intervals, the cleaner answer is that the committee's overall weighting stayed pretty similar after 2019, with at most some limited movement in a few specific metrics.

Project Discussion

Takeaways

Across all four analyses, we found that NCAA tournament seeding is largely explainable by publicly available team metrics, with resume-based measures playing the dominant role. The combined baseline model places 86.3% of teams within two seed lines of the committee's actual decision, with Wins Above Bubble (WAB) being the most influential predictor (its average absolute coefficient is ~1.4 times larger than Barthag and ~6 times larger than the efficiency components). This suggests that the committee primarily seeds teams based on what they accomplished against their schedule, not how strong they look on paper, despite what discourse around tournament seeding often implies.

The remaining ~14% of cases where the model deviates from committee decisions highlight a small but credible set of systematic biases. After controlling for team quality, three power conferences (the SEC, Big East, and Big 12) are consistently over-seeded by 0.3 to 0.6 seed lines, with posterior probabilities of over-seeding above 0.99. These are the only conferences whose 95% credible intervals exclude zero, and the effects survive partial-pooling shrinkage, meaning they are stable rather than the side effects of small-sample noise. On the other hand, neither conference tournament wins or the introduction of the NET ranking system in 2019 appear to meaningfully change committee behavior once team quality is held fixed as we found that per-win effects across all three conference tiers have credible intervals that cross zero, and the no-interaction baseline outperforms both interaction-based NET-era specifications on out-of-sample log-loss.

Overall, our findings suggest that the committee evaluates teams primarily on their full body of work, with a small but visible lean toward power-conferences. The selection process is mostly transparent and reproducible from public data, but not entirely free from identifiable bias.

Limitations

While our project provides a clear framework for evaluating committee behavior, several limitations should be noted. First, all four analyses rely on publicly available metrics, primarily sourced from Barttorvik. The committee may consider information that our predictor set does not capture, such as late-season injuries or end-of-season momentum, and some of the residual variation we attribute to bias may instead reflect these unmeasured variables.

Second, the conference bias analysis estimates a single mean residual per conference and does not model interactions between conference identity and seed range. The SEC over-seeding effect could possibly be concentrated among bubble-tier teams rather than being uniform across all seed lines, which would have very different implications for committee behavior. The conference tournament analysis is similarly constrained by its three-tier grouping, which may not reflect individual conference-level effects.

Lastly, the NET-era analysis is fundamentally limited by sample size as only four post-2019 tournaments are available after excluding 2020, leaving 340 post-NET team-seasons against 738 pre-NET observations. With an imbalance like this, our models have limited power to detect a true shift in committee weighting even if one exists, and the conclusion that "no major shift occurred" should be taken in the context of the specific dataset we are using.

Future Work

There are many possibilities for future work that extends off of our analysis. Firstly, future analyses could incorporate qualitative factors the committee weighs but our metrics do not capture, such as injury status of key players or late-season form indicators, to test whether the residual conference biases shrink once these are accounted for. The conference tournament analysis could be done with conference-specific (rather than tier-specific) random slopes once enough post-NET data accumulates. Likewise, the conference bias analysis could be extended with seed-specific interactions or separate models for top, middle, and bubble seed bands. This would identify whether the over-seeding of power conference teams is concentrated among specific seed tiers. As discussed in our limitations section, as more post-NET seasons become available, that section of the analysis can be revisited with substantially more statistical power.

Overall, this analysis could be paired with bracket outcome models to help quantify the effect of any seeding bias. If an over-seeded conference team faces an easier first-round opponent on average, is there a high chance that they will end up with more wins than their underlying quality would predict? Connecting seeding decisions to tournament outcomes would significantly help in explaining the relationship between committee behavior and impact that their seeding decisions have on tournament teams, and would directly address the influence of any biases identified there.

Github Code Appendix

https://github.com/Danny625/Sports-Analytics-Project